Quaternion Volume: exploration and experiment design

Status and scope

This is a research-design report, not an evidence-review verdict. Novelty is unverified. All initial instances are supplied exact synthetic lattices with planted ambient orders. No finite-field map discovery, performance advantage or ECDLP consequence is established.

Records and implementation

Question RQ-SSI-3e3f54; hypothesis H-SSI-174d36; experiment EXP-SSI-39bae9. Driver tools/quaternion_order_volume.py. Tracking issue https://github.com/aburan28/crypto-autoresearcher/issues/2155. The companion exploration.md contains the derivations, primary-source read receipts, controls and next experiments.

Method and controls

The frozen protocol retains complete-assumption fixtures alongside incomplete bases, nonprimitive generators, misleading partial modules, scalar/collinear maps, and rank/index/closure collisions as applicable. Run integrity replay and software regression checks are distinct from independent mathematical review.

Source and canonical graph policy

Sources and their actual read scopes are recorded in exploration.md and the proposal. Primary pointers: Kani, https://doi.org/10.1007/s13348-010-0029-1; Bisson/Sutherland, https://arxiv.org/abs/0902.4670; Fuselier et al., https://aiezzi.it/documents/art7.pdf; Voight, https://math.dartmouth.edu/~jvoight/quat-book.pdf. The ECDLP frontier map was checked for positioning and left unchanged: this work asserts no ECDLP frontier or scientific-status change. This diagram presents a supplied-object workflow; its dashed edge is an unimplemented research obligation.

Next action

Execute the frozen admitted pilot, retain its observations, and obtain independent review before any conclusion.

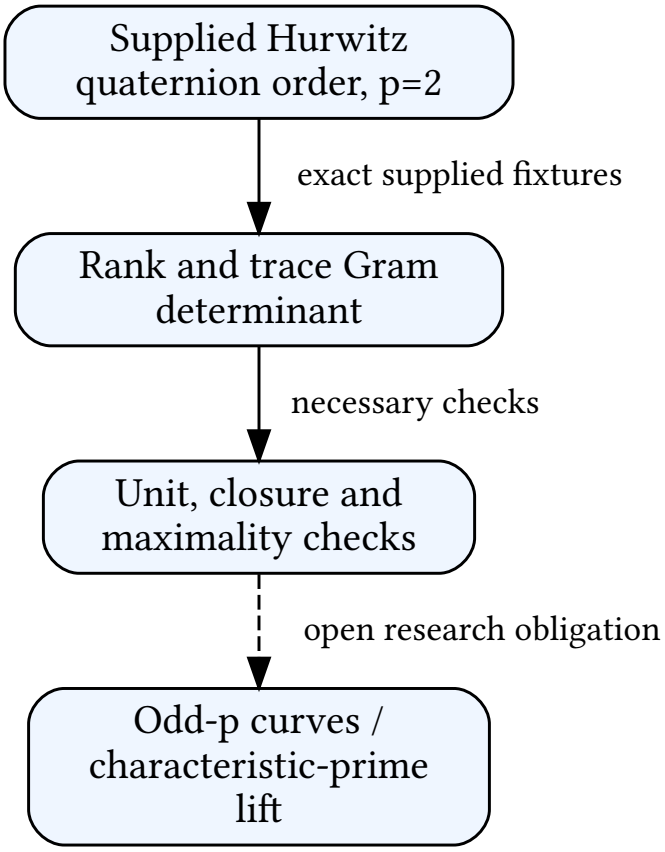

Figure 1: Proposed workflow and explicit unresolved interface.